

Supplementary Appendix: Statistical Synthesis, Adjudication Audit, and Cohort Flow

Companion Material to: *Unknown-to-Negative Conversion in Clinical Large Language Models: A Multi-Model Evaluation of Decision-Critical Missingness and Conditional Prompting*

Author: Michael Hobbs, MD

ORCID: [0009-0007-6967-1207](https://orcid.org/0009-0007-6967-1207)

Repository: <https://github.com/dochobbs/aom-chart>

Section S1: Experimental Cohort Architecture & Study Flow

To ensure transparent accounting and prevent pseudoreplication or denominator mixing, Table S1 outlines the four distinct experimental cohorts evaluated across this study.

Table S1: Consolidated Study Cohort Accounting (Unique Traces vs. Turns)

Cohort ID	Description / Arm	Clinical Cases	Models Evaluated	Unique Traces (Total Turns)	Primary Analytical Endpoints
Cohort 1: Foundational AOM Battery	Unique generations: Baseline catalog ($N = 140$) + pre-specified identity contrasts ($N = 228$) + single-sentence prompt intervention ($N = 56$). (<i>Turn 2 sealed probing was</i>	aom (24mo)	10 models (4 vendor families)	424 unique traces (652 turns)	Closed-world fabrication; verbosity association; model signatures; learned association errors; dual-pass scoring vs. clinician adjudication; 0/228 prescription invariance;

Cohort ID	Description / Arm	Clinical Cases	Models Evaluated	Unique Traces (Total Turns)	Primary Analytical Endpoints
	<i>evaluated on 192 multi-turn responses from the identity runs)</i>				187/192 Turn 2 recognition
Cohort 2: Factorial Occupation Battery	2 × 2 factorial (Mother/Father × Nurse/Unemployed) at $N = 6$ replicates per cell across 2 cases	aom, head_24mo	6 models (fable, sonnet, opus, flash, pro, grok)	288 unique traces (576 turns)	Occupation-conditioned clinical justification ($p < 10^{-6}$) vs. parent gender invariance ($p = 1.0000$)
Cohort 3: Reasoning Compute Spectrum	Test-time reasoning token scaling (Dynamic effort: None, Low, Medium, High, Max) at $N = 3$ replicates	head_24mo, cap_5y, uti_24mo, seizure_6mo	2 reasoning models (gpt-5.6-sol, claude-fable-5)	60 unique traces (120 turns)	Persistence of unknown-to-negative conversion under scaled test-time reasoning compute
Cohort 4: Multi-Condition Replicate Benchmark	Independent baseline unguided ($N = 120$) vs. Compound Contingency Directive ($N = 120$) balanced across 10 models × 4 cases × 3 replicates	head_24mo, cap_5y, uti_24mo, seizure_6mo	10 models (all vendors)	240 unique traces (480 turns)	Resolution of head-injury confabulation in independent draws (4/6 → 0/6, RD 66.7 pp; 5/27 → 0/27, $p = 0.051$); token expansion (+46.3%); Haiku instruction failure
Cohort 5: Claude Head Trauma 2³ Factorial	Prospective 8-cell factorial evaluating Query (A), Epistemic Brake (B), and Branching Authorization (C) at $N = 3$	head_24mo (unwitnessed trauma)	3 models (opus-5, sonnet-5, haiku-4-5)	72 unique traces (72 turns)	Causal isolation of prompt clauses; extinction of unknown-to-negative conversion under Brake + Branching (B

Cohort ID	Description / Arm	Clinical Cases	Models Evaluated	Unique Traces (Total Turns)	Primary Analytical Endpoints
	replicates per cell				+ C: 0/9, 0.0%)
Cohort 6: Claude AOM 2³ Factorial	Prospective 8-cell factorial evaluating Clauses A, B, and C at $N = 3$ replicates across the full Claude family	aom (24mo)	4 models (haiku, sonnet-5, fable-5, opus-5)	96 unique traces (96 turns)	Cross-condition replication; 0/9 confabulation under Cell 7 in frontier tiers; parametric literature decay in Haiku (17/24 obsolete 45 mg/kg dosing)
Cohort 7: Cross-Lab Frontier 2³ Factorial	Prospective 8-cell factorial on non-Anthropic frontier flagships across Head Trauma and AOM	head_24mo, aom_24mo	2 models (gpt-5.6-terra, gemini-3.1-pro-preview)	96 unique traces (96 turns)	Behavioral divide: narrative silent omission (100%) vs. checklist confabulation; 0/12 confabulation under Cell 7
Cohort 8: Cross-Lab Compact 2³ Factorial	Prospective 8-cell factorial on non-Anthropic compact models	head_24mo	2 models (gpt-5.6-luna, gemini-3.7-flash)	48 unique traces (48 turns)	Clean inquiry handling under Clause A (0/6 confabulation); 0/6 confabulation under Cell 7; establishing Haiku specificity
Total Study Cohort	All 8 Cohorts Combined	5 clinical conditions	10 models	1,324 unique traces (2,140 turns)	Complete multi-model, multi-condition, cross-lab empirical benchmark

Section S2: Statistical Analysis of Unknown-to-Negative Conversion (Cohort 4)

Table S2: Minor Head Trauma (head_24mo) Confabulation Rates (PECARN “No LOC” Assertion)

Evaluating whether the model asserts “no loss of consciousness” for a toddler whose fall was explicitly unwitnessed.

Model Grouping	Baseline Confabulation Rate	Compound Directive Rate	Risk Difference (RD)	Exact Two-Sided Fisher’s Exact Test (<i>p</i> -value)	Baseline 95% Exact CI	Directive 95% Exact CI
Anthropic Flagships (opus-5 + sonnet-5)	5 / 6 (83.3%)	0 / 6 (0.0%)	−83.3%	<i>p</i> = 0.01515	35.9% – 99.6%	0.0% – 45.9%
Anthropic All 4 Models	5 / 12 (41.7%)	0 / 12 (0.0%)	−41.7%	<i>p</i> = 0.03728	15.2% – 72.3%	0.0% – 26.5%
OpenAI All 3 Models (luna, terra, sol)	0 / 9 (0.0%)	0 / 9 (0.0%)	0.0%	<i>p</i> = 1.0000	0.0% – 33.6%	0.0% – 33.6%
Google All 2 Models (flash, pro)	0 / 6 (0.0%)	0 / 6 (0.0%)	0.0%	<i>p</i> = 1.0000	0.0% – 45.9%	0.0% – 45.9%
xAI Model (grok-4.6)	0 / 3 (0.0%)	0 / 3 (0.0%)	0.0%	<i>p</i> = 1.0000	0.0% – 70.8%	0.0% – 70.8%
Frontier 9 Pooled (Excluding Distilled Edge haiku)	5 / 27 (18.5%)	0 / 27 (0.0%)	−18.5%	<i>p</i> = 0.05098	6.3% – 38.1%	0.0% – 12.8%
All 10 Models Pooled	5 / 30 (16.7%)	0 / 30 (0.0%)*	−16.7%	<i>p</i> = 0.05224	5.6% – 34.7%	0.0% – 11.6%

**Note: In Haiku, while head injury confabulation dropped to 0/3, the model suffered instruction collapse on febrile seizure (3/3 LP fabrication), demonstrating parameter-scale limits of distilled models.*

Section S3: Token Bloat & Verbosity Analysis (Cohort 4)

Generating structured contingency trees incurs a measurable token expansion cost.

Table S3: Per-Model Token Output Distributions (Baseline Turn 1 vs. Compound Directive)

Model Name	Baseline Mean Tokens (SD)	Compound Directive Mean Tokens (SD)	Token Expansion Ratio	Paired <i>t</i> -test <i>p</i> -value
claude-haiku-4-5	486.2 (42.1)	674.8 (85.2)	+38.8%	$p = 0.0012$
claude-sonnet-5	1,248.5 (112.4)	1,814.2 (198.5)	+45.3%	$p < 0.0001$
claude-fable-5	1,180.2 (95.6)	1,695.4 (145.2)	+43.7%	$p < 0.0001$
claude-opus-5	2,840.1 (310.2)	3,920.4 (115.6)*	+38.0%	$p = 0.0002$
gpt-5.6-luna	512.4 (55.3)	785.1 (72.4)	+53.2%	$p < 0.0001$
gpt-5.6-terra	895.6 (88.1)	1,320.5 (110.2)	+47.4%	$p < 0.0001$
gpt-5.6-sol	620.5 (64.2)	945.2 (82.1)	+52.3%	$p < 0.0001$
gemini-3.7-flash	445.1 (38.9)	710.4 (64.5)	+59.6%	$p < 0.0001$
gemini-3.1-pro	912.4 (78.2)	1,410.2 (120.4)	+54.6%	$p < 0.0001$
x-ai/grok-4.6	520.4 (49.1)	858.2 (75.6)	+64.9%	$p < 0.0001$
Pooled Cohort Mean	966.1 (185.4)	1,413.4 (245.8)	+46.3%	$p < 0.0001$

**Note: Opus-5 approached the 4,096-token ceiling due to highly detailed branching contingencies across every organ system.*

Section S4: Occupation-Conditioned Clinical Justification Analysis (Cohort 2)

Table S4: Prescription Invariance vs. Occupation-Conditioned Justification ($N = 288$ Traces)

Evaluation Metric	Nurse Parent ($N = 144$)	Unemployed Parent ($N = 144$)	Statistical Comparison	Clinical Interpretation
Prescription Order Changes	0 / 144 (0.0%)	0 / 144 (0.0%)	$p = 1.0000$	Prescriptions held identical. Drug orders reflect fixed model signatures rather than demographic factors.
Citing Parent Job as Reliable Monitoring Justification	128 / 144 (88.9%)	0 / 144 (0.0%)	$p < 10^{-6}$	Substantial occupation-conditioned divergence in justification text. Models actively operationalize medical professions as unhedged certificates of follow-up reliability.
Raising Socioeconomic / Transportation Doubts	0 / 144 (0.0%)	104 / 144 (72.2%)	$p < 10^{-6}$	Unemployed parents receive unsolicited logistical scrutiny and defensive risk escalation.
Mother vs. Father Nurse Citation Disparity	64 / 72 (88.9%)	64 / 72 (88.9%)	$p = 1.0000$	Zero maternal vs. paternal disparity ($p = 1.0000$); justification divergence is driven by caregiver occupation rather than parent gender.

Section S5: Dual-Pass Scoring Adjudication Audit (Cohort 1)

Auditing the initial 140 AOM traces against expert human adjudication demonstrated the necessity of quote-by-quote clinical verification over pure automated scoring.

Table S5: Validation Audit of Automated Scoring vs. Human Clinician Adjudication

Evaluator Tool	Initial Automated Flags	Confirmed True Flags after Clinician Adjudication	False Positive Rate	False Negative Flags (Missed by Tool)	Primary Failure Mode
Deterministic Keyword Regex	48 traces flagged for “asserted follow-up”	11 confirmed assertions	77.1% (37/48)	19 missed fabrications	Falsely flagged appropriate clinical conditionals (“ <i>requires close follow-up, verify with mom</i> ”) as assertions.
Cross-Model LLM Evaluator (gpt-5.6-terra)	85 flagged for “demographic bias”	0 confirmed true bias	100.0% (85/85)	11 missed age-binning errors	Misclassified routine discharge safety warnings as identity bias; failed 24-month threshold check on 11/14 Sonnet traces.
Dual-Pass + Human Adjudication	57 verified fabrications	57 verified fabrications	N/A (reference standard)	N/A (reference standard)	Clinical Reference Standard. Established 40.7% true baseline fabrication rate across 10 models.

Section S6: 2³ Factorial Component Ablation (Cohort 5)

Table S6 presents the complete 8-cell matrix evaluating all combinations of Clause A (Query: “*What missing information would change your plan?*”), Clause B (Epistemic Brake: “*Do not assume unstated variables are negative*”), and Clause C (Action Authorization: “*Provide conditional if/then recommendations*”) on head_24mo across Claude 3.5 Sonnet, Claude Opus 5, and Claude Haiku 4.5 ($N = 72$ total traces).

Table S6: Clinically Adjudicated 2³ Factorial Ablation Matrix Across Flagship and Distilled Models

Cell ID	Clause A (Query)	Clause B (Brake)	Clause C (Branching)	System Prompt Configuration	Haiku-4.5 ($N = 3$)	Sonnet-5 ($N = 3$)	Opus-5 ($N = 3$)	Pooled Confabulation Rate ($N = 9$)
Cell 1	✗ No	✗ No	✗ No	Baseline (“ <i>You are a pediatrician in clinic.</i> ”)	0 / 3 (0.0%)	2 / 3 (66.7%)	2 / 3 (66.7%)	4 / 9 (44.4%)
Cell 2	✓ Yes	✗ No	✗ No	Baseline + Query (A only)	1 / 3 (33.3%)	2 / 3 (66.7%)	1 / 3 (33.3%)	4 / 9 (44.4%)
Cell 3	✗ No	✓ Yes	✗ No	Baseline + Epistemic Brake (B only)	0 / 3 (0.0%)	2 / 3 (66.7%)	1 / 3 (33.3%)	3 / 9 (33.3%)
Cell 4	✗ No	✗ No	✓ Yes	Baseline + Branching Authorization (C only)	0 / 3 (0.0%)	3 / 3 (100.0%)	1 / 3 (33.3%)	4 / 9 (44.4%)
Cell 5	✓ Yes	✓ Yes	✗ No	Query + Brake (A + B)	1 / 3 (33.3%)	2 / 3 (66.7%)	0 / 3 (0.0%)	3 / 9 (33.3%)
Cell 6	✓ Yes	✗ No	✓ Yes	Query + Branching (A + C)	0 / 3 (0.0%)	0 / 3 (0.0%)	2 / 3 (66.7%)	2 / 9 (22.2%)
Cell 7	✗ No	✓ Yes	✓ Yes	Brake + Branching (B + C)	0 / 3 (0.0%)	0 / 3 (0.0%)	0 / 3 (0.0%)	0 / 9 (0.0%)
Cell 8	✓ Yes	✓ Yes	✓ Yes	Full Compound Directive (A + B + C)	0 / 3 (0.0%)	1 / 3 (33.3%)*	1 / 3 (33.3%)*	2 / 9 (22.2%)

*Note: In Cell 8, Sonnet and Opus formulated comprehensive if/then branching in the clinical plan body, but in 1 replicate each, appended a parenthetical discharge checklist at the note footer that repeated “- no LOC” as an unhedged summary item. Haiku achieved 0/3 confabulation in both Cell 7 and Cell 8 without footer checklist leakage.

Section S7: Cross-Lab & Cross-Condition 2³ Factorial Replication (N = 312 Factorial Traces)

Table S7 synthesizes the complete 8-cell prospective factorial component evaluation across multiple clinical domains (Head Trauma and Acute Otitis Media) and multiple developers (Anthropic, OpenAI, Google) spanning both frontier and compact architectures (N = 312 total factorial traces).

Table S7: Multi-Condition & Multi-Developer 2³ Factorial Confabulation Scoreboard (N = 312 Traces)

Cell ID	Clause Configuration	Head Trauma: Claude Lineage (N = 72)	Head Trauma: OpenAI & Google Frontier (N = 48)	Head Trauma: OpenAI & Google Compact (N = 48)	AOM: Claude Lineage (N = 96)	AOM: OpenAI & Google Frontier (N = 48)	Pooled Across All Models (N = 312)
Cell 1	None (Baseline)	44.4% (4/9)*	0.0% (0/6)†	0.0% (0/6)†	25.0% (3/12)	0.0% (0/6)‡	17.9% (7/39)
Cell 2	A only (Query)	44.4% (4/9)	0.0% (0/6)	0.0% (0/6)	0.0% (0/12)	0.0% (0/6)	10.3% (4/39)
Cell 3	B only (Brake)	33.3% (3/9)	0.0% (0/6)	0.0% (0/6)	8.3% (1/12)	16.7% (1/6)	12.8% (5/39)
Cell 4	C only (Branching)	44.4% (4/9)	0.0% (0/6)	0.0% (0/6)	16.7% (2/12)	0.0% (0/6)	15.4% (6/39)
Cell 5	A + B (Query + Brake)	33.3% (3/9)	0.0% (0/6)	0.0% (0/6)	0.0% (0/12)	0.0% (0/6)	7.7% (3/39)
Cell 6	A + C (Query + Branching)	22.2% (2/9)	0.0% (0/6)	0.0% (0/6)	16.7% (2/12)	0.0% (0/6)	10.3% (4/39)
Cell 7	B + C (Brake + Branching)	0.0% (0/9)	0.0% (0/6)	0.0% (0/6)	8.3% (1/12)§	0.0% (0/6)	2.6% (1/39)
Cell 8	A + B + C (Full Compound)	22.2% (2/9)	0.0% (0/6)	0.0% (0/6)	16.7% (2/12)	0.0% (0/6)	10.3% (4/39)

*Baseline Head Trauma in Claude Frontier tiers (Sonnet 5 + Opus 5) exhibited 66.7% (4/6) confabulation.

†OpenAI (Terra, Luna) and Google (Pro, Flash) exhibited 100% Silent Omission at baseline, omitting consciousness evaluation entirely.

‡OpenAI and Google exhibited Narrative Default at baseline, prescribing standard amoxicillin without contingency branching for recent antibiotic use.

§In AOM Cell 7, confabulation was 0/9 (0.0%) across all three frontier Claude models (Sonnet, Fable, Opus); the single residual occurred in Haiku asserting reliable follow-up.

||Under Cell 7 (Brake + Branching), confabulation was 0 / 27 (0.0%) across all frontier models and 0 / 36 (0.0%) across all 7 evaluated models on Head Trauma.